

BACS2103 Software Quality Assurance and Testing – Assignment

1. Objectives

This assignment provides students with the opportunity to learn and apply the following skills;

- Creating a test plan and test cases.
- Perform appropriate testing techniques.

2. Assessment

The learning outcome assessed is:

CLO2: Adopt software quality standard in the development of software test plan and test cases. (A3, PLO4)

Contribution of marks to coursework:

- The assignment deliverables* contributes **60%** (under CLO2) to the coursework component as follows:

Parts	Marks
Create Test Plan and Test Cases	30
Findings Robotic Process Automation (RPA) Tool	20
Apply standard ISO/IEC/IEEE 29119-3 in Test Plan and Test Cases	30
Individual Collaboration and Team Cooperation	20
Total	100

** Note: Tutor has the right to adjust the marks based on the % contribution by each team member.*

Refer to the Assignment Rubrics for details.

3. Group Formation

Students will be assigned to groups of **4 - 5** members per team from the same tutorial group by the tutor.

Every team member is expected to contribute and participate actively in the entire process of completing the assignment to develop complete documentation by creating a test plan, creating test cases, and performing appropriate testing techniques. Sharing of ideas and assistance in the completion of assignment among members is required.

4. Assignment Tasks

Students need to select ONE (1) published and currently functioning system, web, or mobile application as a sample to be tested and to create test documentations by applying software testing standard ISO/IEC/IEEE 29119-3 in the test documentation. As part of the testing environment, students are required to identify one Robotic Process Automation (RPA) tool which will be proposed to support the testing activities by studying on the features of the selected tool.

Students may not need to include ALL topics/sections in **ISO/IEC/IEEE 29119-3** for the report but the following topics **MUST** be included:

1. Test Plan
2. Test Case (Appendix - FORM 1)
3. Test Execution Status Summary

5. Practical Schedule

Week	Practical Task	Deliverable
1	Discussion on assignment question, assignment task, and assignment group formation	Week 1's task(s)
2	Discussion on the selected system, web or mobile application for assignment. List the individual modules to be tested	Week 2's task(s)
3	Discussion on the implementation of ISO/IEC/IEEE 29119-3 standard for Test Plan, Test Case Specification, Test Execution Status Summary preparation	Week 3's task(s)
4	Discussion on the context of testing of the Test Plan	Week 4's task(s)
5	Discussion on the risk register of the Test Plan	Week 5's task(s)
6	Discussion on the test strategy of the Test Plan – test design technique(s)	Week 6's task(s)
7	Discussion on the test strategy of the Test Plan – test entry and exit criteria	Week 7's task(s)
8	Discussion on the test strategy of the Test Plan – test environment, RPA testing tool, test data	Week 8's task(s)
9	Discussion on the testing activities and estimates of the Test Plan	Week 9's task(s)
10	Discussion on the staffing and schedule of the Test Plan	Week 10's task(s)
11	Discussion on Test Case Specification	Week 11's task(s)
12	Discussion on Test Execution Status Summary	Week 12's task(s)
13	Review on Test Plan, Test Case Specification, Test Execution Status Summary. Assignment Submission	Week 13's task(s)
14	Feedback on the assignment	Week 14's task(s)

6. The Final Assignment Report Format

The final assignment report for all Parts should contain the following items:

- Cover sheet (Appendix - FORM 2)
- Plagiarism statement with student signatures (Appendix - FORM 3)
- Assessment Criteria (Group and Individual)
- Table of Contents
- Body of answers
- Appendices (if any)

The report must be type-written using **MS-Word**. You are recommended to format your report according to the following specification:

Media	Softcopy - well written and properly formatted report to be submitted via uploading softcopy to Google Classroom which include all work/deliverables for Assignment report.
Font Size	A body text of font <i>size 12</i> is required while for headings and subheadings a larger font size must be used.
Font Style	Use <i>Times New Roman</i> for body text. Main headings and sub-headings should be clearly stated using suitable font styles (e.g. Arial).
Line Spacing	Typed material should be <i>1.5-line spaced</i> .
Alignment	Use <i>Justify</i> for alignment.
Headers and Footers	Appropriate footers and headers should be used to enhance clarity and presentation.
Page Numbering	Ensure that all pages (except cover page) are numbered.
Paper Size	Use <i>A4</i> paper (29.7cm x 21cm).

Table 1: Written Report Format

7. Late Submission

All assignments should be submitted by **Week 13, Friday, 11.59 pm**.

Late submission of the assignment will be handled according to the Guideline for Late Submission of Coursework available at **TAR UMT's Intranet Examinations and Credit Accumulation Undergraduate Guideline**.

PART B : LATE SUBMISSION OF COURSEWORK INFORMATION

Please check/tick one of the information below :

☐	Late submission of 1 - 3 days after deadline of submission: minus 10 marks
☐	Late submission of 4 - 7 days after deadline of submission: minus 20 marks
☐	Late submission of > 7 days after deadline of submission: 0 mark

In certain circumstances, a student may be allowed to submit the assignment late with valid reason. S/he must inform the respective tutor **at least one week before** the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late, but **no guarantee** that the students will not be penalized.

As a general rule, no extension of time will be granted. The assignment question and its due dates are normally disclosed in advance to students in order that they will be able to manage their time according to different course study progress and complete this assignment on time.

8. Academic Integrity and Plagiarism

Students are required to redo the entire assignment if the contents of any part of the assignment does not comply with the requirements of the assignment as requested by the respective lecturer / tutor.

Before submitting your assignment, please make sure that you have complied with **TAR UMT Plagiarism Policy**. Any cheating, attempt to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment will cause the students concerned to be penalized.

IMPORTANT: Students found to be dishonest are liable to disciplinary action.

TEST CASE TEMPLATE

Test Case Template	
Test Case #:	Test Case Name:
System:	Subsystem:
Design By:	Design Date:
Executed By:	Execution Date:
Short Description:	

Pre-conditions:

Step	Action	Expected System Response	Pass/Fail	Comments
1				
2				
3				
4				
5				

Post-conditions:



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Programme: _____ (Group: _____)

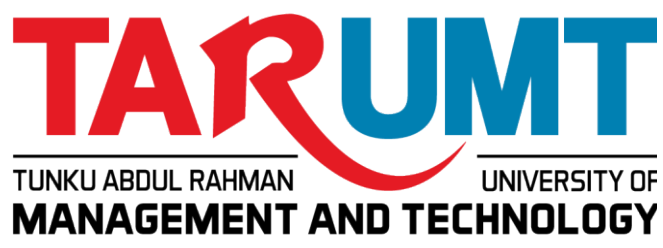
Assignment

BACS2103 SOFTWARE QUALITY ASSURANCE AND TESTING

Name (Block Letters)	Registration No.	Signature	Marks

Tutor's Name: _____

Date of Submission: _____



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Plagiarism Statement and Guideline for Late Submission of Coursework

Read, complete, and sign this statement to be submitted with the written report.

We confirm that the submitted work is all our own work and are in our own words.

Name (Block Letters)	Registration No.	Signature & Date	Picture